Explosion Protection

Introduction and Basics



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Overview

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2. **Legal Basics – Manufacturer**
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1. Physical Basics

Explosions: Requirements and Impacts



Hemel Hempstead (England), December 2005

Tank farm:

- A storage tank was filling with petrol
- Both level gauge and independent high-level switch failed
  - → *Overflow of approx. 190 t petrol (approx. 250 m^3) and eventually formation of an explosive atmosphere*
  - → *Subsequent explosions and major fire*
  - → *Considerable damage to property, the environment and persons*



Chiltern Air Support Unit and Hertfordshire County Council

Effects of Explosions

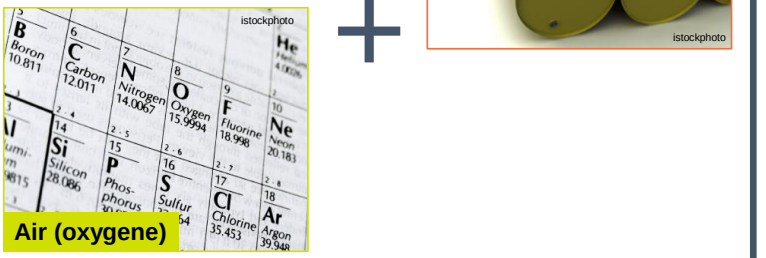


Chiltern Air Support Unit and Hertfordshire County Council

Effects of explosions are – among other things:

- Damage to persons (e.g. pressure, flames)
- Environmental contamination (e.g. emission of toxic substances in the air or water)
- Damage to property (e.g. destruction of plants, buildings)
- Financial loss (e.g. production downtimes, monetary fines)
- Damage to image (e.g. media coverage)

Requirements for an Explosion



Explosive atmosphere

Term: Explosive Atmosphere

Mixture with air, under atmospheric conditions, of flammable substances in the form of gases, vapours, mists or dusts in which, after ignition has occurred, combustion spreads to the entire unburned mixture.

Directive 2014/34/EU; article 2.4

Directive 1999/92/EC; article 2

Term

Citation

Literature

Atmospheric Conditions



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Atmospheric conditions are referred to as:

- A temperature range from -20 … +60 °C
- A pressure range from 800 ... 1.100 hPa (0,8 … 1,1 bar)
- Oxidiser: air (oxygen concentration 21 vol.-%)

→ *Under these atmospheric conditions material safety data sheets, standards on types of protection etc. are applicable*

2. Legal Basics – Manufacturer

Directive 2014/34/EU



Treaty on Functioning of the European Union

Article 114: free movement of goods		Article 153: occupational health and safety
directive 2014/34/EU „manufacturer's directive" on products		**directive 1999/92/EC** „users' directive" on safety at work
***Literal* transposition to national law** ▪ Groups and categories ▪ Conformity assessment ▪ Marking ▪ Documentation ▪ …		**Transposition of *minimum requirements*** ▪ Priority of protective measures ▪ Zone classification of hazardous areas ▪ Explosion protection document ▪ Inspection ▪ …
Comparable directives for: ▪ Machinery 2006/42/EC ▪ Pressure equipment 97/23/EC; 2014/68/EU ▪ …		**Comparable directives for:** ▪ Use of work equipment directive 2009/104/EC ▪ …

Groups and Categories

Group I: mining applications		Group II: other applications		
classified into:		*classified into:*		
Category M1	Category M2	Category 1	Category 2	Category 3
equipment ensures:		*equipment ensures:*		
Very high level of protection	High level of protection	Very high level of protection	High level of protection	Normal level of protection
marked:				
Ex I M1	Ex I M2	Ex II 1 G II 1 D	Ex II 2 G II 2 D	Ex II 3 G II 3 D

G: hazardous gas atmospheres
D: hazardous dust atmospheres

Conformity Assessment Procedures

Category 1		Category 2		Category 3
All products		Combustion engines, Electrical devices	Other products	All products
EU-type examination (annex III)	EU-type examination (annex III)	EU-type examination (annex III)	EU-type examination (annex III)	Internal control of production (annex VIII)
Production quality assurance (annex IV)	Product verification (annex V)	Conformity to type (annex VI)	Product quality assurance (annex VII)	Communication of documents

Alternative: unit verification (annex IX)

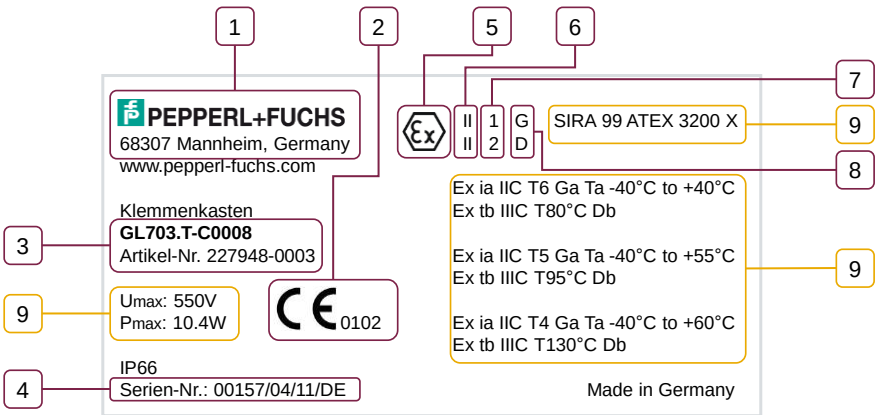
Marking Requirements acc. to 2014/34/EU

Legal marking acc. to 2014/34/EU

1	name and address of the manufacturer
2	CE marking (exception: components)
3	Designation of series or type
4	Serial number (if any) or year of construction
5	Symbol ⟨Ex⟩
6	Group (I or II)
7	Category (for group II: 1, 2 or 3)
8	For group II: letter „G“ and/or „D“
9	Where necessary: all information essential to their safe use

Technical marking (e.g. acc. to EN 60079-...)

Example: Label of a Junction Box



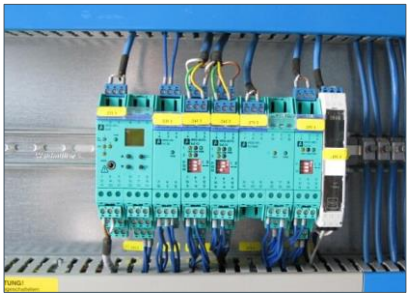
Legal Marking: Group and Category



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Luminaire in hazardous gas areas:

II 2 G



Peter Leitenmaier, TÜV SÜD Industrie Service

Intrinsically safe barrier:

II (1) G



Pepperl+Fuchs

Level sensor with multiple categories:

II 1/2 G

3. Legal Basics – User

Directive 1999/92/EC



Treaty on Functioning of the European Union

Article 114:
free movement of goods

directive 2014/34/EU
„manufacturer's directive" on products

***Literal* transposition to national law**

- Groups and categories
- Conformity assessment
- Marking
- Documentation
- …

Comparable directives for:

- Machinery 2006/42/EC
- Pressure equipment 97/23/EC; 2014/68/EU
- …

Article 153:
occupational health and safety

directive 1999/92/EC
„users' directive" on safety at work

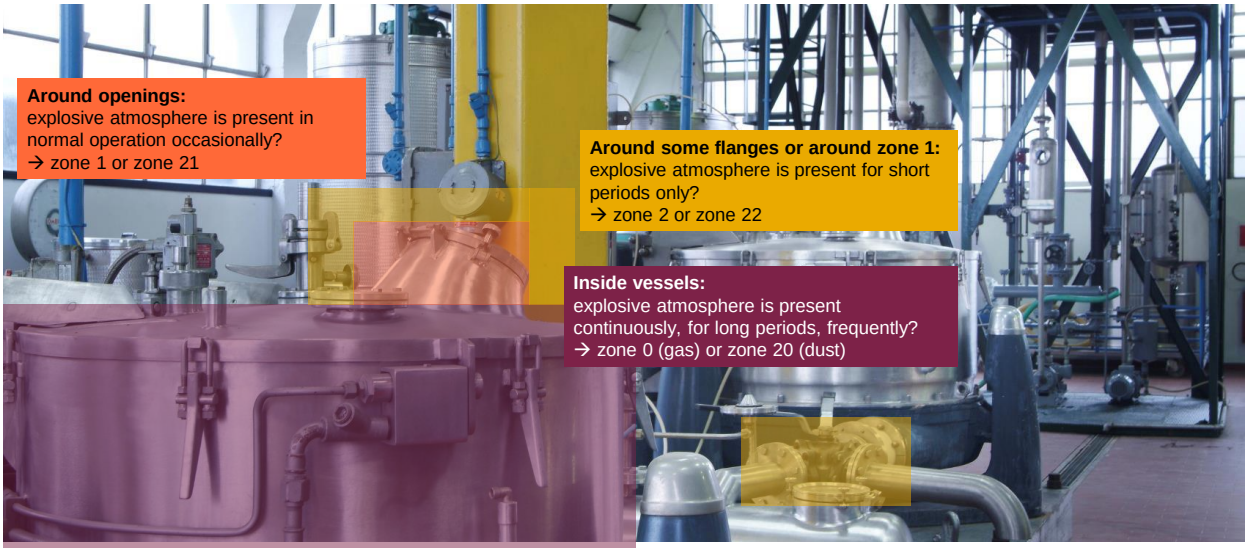
Transposition of *minimum requirements*

- Priority of protective measures
- Zone classification of hazardous areas
- Explosion protection document
- Inspection
- …

Comparable directives for:

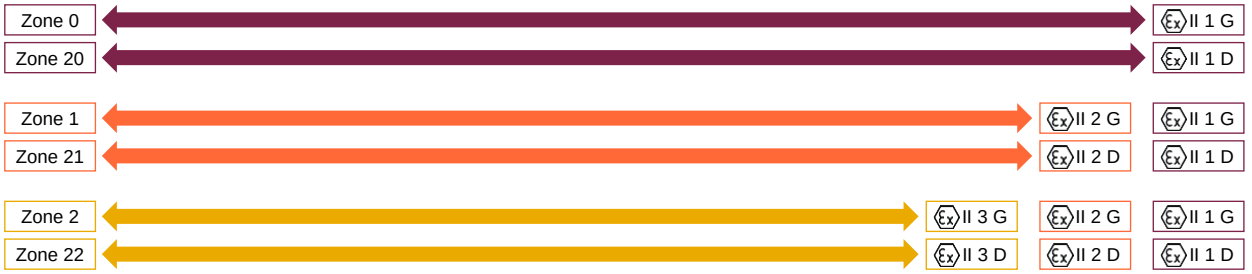
- Use of work equipment directive 2009/104/EC
- …

Classification of Hazardous Areas



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Zones and Categories



Assignment of the equipment category to the respective zone:

- Is often carried out as indicated above, unless otherwise specified in the explosion protection document
- Is only applicable if the formation of an explosive atmospheric and the presence of an ignition source are independent of each other
  - → *A risk assessment is to be carried out*

4. Technical Basics

Standards on Explosion Protection



Standards on Explosion Protection



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National laws (often based on EU directives) are legally binding:

- Describe safety objectives
- Do normally not include specific technical solutions
  - → *Application of European or national standards can be helpful*

Example: Development of Standards

2013

International: Electrical installations design, selection and erection
- IEC 60079-14 (Edition 5.0; 2013-11-28)
  - *New: IEC 31J/301/CD:2019*

2014

Europe:
- EN 60079-14:2014

National Editions UK, Switzerland, Germany:
- BS EN 60079-14:2014-06-30
- SN EN 60079-14:2014-03
- DIN EN 60079-14 (VDE 0165-1:2014-10):2014-10

etc.

Other Member States
- Etc.

European Standards

Manufacturer	User
EN 1127-1 (concepts and methodology)	EN 1127-1 (concepts and methodology)
EN 60079-0 (equipment: general requirements)	EN 60079-10-1 (zone classification - gas)
EN 60079-1 (flameproof enclosure “d”)	EN 60079-10-2 (zone classification - dust)
EN 60079-7 (increased safety “e”)	EN 60079-14 (design, selection, installation)
EN 60079-11 (intrinsic safety “i”)	EN 60079-17 (inspection)
EN 60079-15 (type of protection “n”)	EN 60079-19 (repair and overhaul)
EN 60079-18 (encapsulation “m”)	EN 60079-29-2 (gas detectors)
EN 60079-28 (optical radiation “op”)	EN 60079-30-2 (resistance trace heatings)
EN 60079-29-1 (gas detectors)	…
EN 60079-30-1 (resistance trace heatings)	
EN 60079-31 (protection by enclosure “t”)	
…	

5. Organisational Basics

Qualification of Personnel



Working in Hazardous Areas



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Different activities in hazardous areas require different qualification:

- Design, equipment selection, installation
- Inspection and maintenance
- Repair and overhaul
  - → *Qualification requirements may be regulated by national law*

Qualification: Laws and Standards

Legal requirements?		Requirements acc. to EN 60079-…
?	Design	EN 60079-14, annex A for - Responsible persons - Operatives - Designers
	Selection	
	Erection	
?	Inspection and Maintenance	EN 60079-17
?	Repair and Overhaul	EN 60079-19

EN 60079-14, Annex A

Responsible person	Operatives	Designer
▪ Management of operatives	▪ Selection and erection	▪ Design and selection
▪ General understanding of relevant electrical engineering ▪ Read and assess engineering drawings ▪ Practical understanding of explosion protection principles ▪ Working knowledge and understanding of relevant standards ▪ Basic knowledge of quality assurance (incl. auditing, documentation, traceability of measuring and calibration)	▪ General principles of explosion protection ▪ Principles of the types of protection and marking ▪ Certificates and standards ▪ General understanding of the inspection requirements acc. to IEC/EN 60079-17 ▪ Work permit systems and safe isolation	▪ Detailed knowledge of the general principles of explosion protection ▪ Detailed knowledge of the principles of the types of protection and marking ▪ Detailed knowledge of certificates and standards ▪ Detailed knowledge of work permit systems and safe isolation ▪ General understanding of the inspection requirements acc. to IEC/EN 60079-14
▪ Proof of competences at regular intervals		

Qualification of Personnel

The design of the installation, the selection of equipment and the erection covered by this standard shall be carried out only by persons whose training has included instruction on the various types of protection and installation practices, relevant rules and regulations and on the general principles of area classification. […] (see Annex A).

Appropriate continuing education or training shall be undertaken by personnel on a regular basis.

NOTE Competency can be demonstrated in accordance with a training and assessment framework relevant to national regulations or standards or user requirements

EN 60079-14:2014, 4.5

Term

Citation

Literature